

A Filippov system describing media effects on the spread of infectious diseases

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ABSTRACT

A Filippov epidemic model with media coverage is proposed to describe the real characteristics of media/psychological impact in the spread of an infectious disease. We extend the existing models by incorporating a piecewise continuous transmission rate to describe that the media coverage exhibits its effect once the number of infected individuals exceeds a certain critical level. Mathematical and bifurcation analyses with regard to the local, global stability of equilibria and local sliding bifurcations are performed. Our main results show that the system stabilizes at either the equilibrium points of the two subsystems or the new endemic state induced by the on–off media effect, depending on the threshold levels. The finding suggests that a previously chosen level of the desired number of infected individuals can be reached when the threshold policy and other parameters are chosen properly.

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1. Introduction

When a type of contagious disease appears and starts to spread, people's response to the threat of disease is dependent on their perception of risk, which is affected by public and private information disseminated widely by the media. Massive news coverage and fast information flow can generate a profound psychological impact on the public. Media communications have played an important role in affecting the outcome of infectious disease outbreaks, such as the 2003 severe acute respiratory syndrome (SARS) and the recent 2009 H1N1 influenza epidemic [1–6]. In particular, the propagation of information and education by media have great influence not only on individual behavior but also on the implementation of public intervention and control measures. If susceptible individuals are alert enough to avoid unnecessary contact with infected individuals or all infected individuals stay in the region being treated and cancel their trips, new transmission will apparently decrease due to reduced contact rates [7,8]. The role of media coverage on the transmission of disease is thus crucial and should be given prominence in the study of disease dynamics.

There has been an increasing interest in models in which the impact of media coverage on disease spread in a given population is considered. Liu et al., in [1], described the impact of media coverage on the transmission coefficient by a decreasing factor, resulting in the transmission coefficient $\beta \exp(-a_1 E - a_2 I - a_3 H)$, where E , I , and H are the numbers of reported exposed, infectious, and hospitalized individuals, respectively. Cui et al., in [3], proposed a compartment model with incidence rate $\mu \exp(-mI)SI$ with $m > 0$ to reflect the impact of media coverage on the transmission. They have shown that Hopf bifurcation can occur when $m > 0$ is sufficiently small. Li et al., in [6], used $\beta_2 I/(m + I)$ to reflect the reduced amount of contact rate through media coverage and formulated an SIS (susceptible–infective–susceptible) epidemic model with incidence rate $(\beta_1 - \beta_2 I/(m + I))SI/N$. Moreover, other forms, such as $(\mu_1 - \mu_2 f(I))SI/(S + I)$, have been formulated to approximate the media-induced incidence rate (see details in [2,4,5]).

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A common assumption for the models in [1–6] is that the impact of media coverage on the transmission of the infectious disease occurs as soon as the disease emerges and remains during the whole process of the disease's spreading. However, this is not really the case. Mostly, at the initial stages of an emerging infectious disease, both the general individuals and public mass media are unaware of the disease. Media reports, information processing, and individuals' alerted responses to the information can only arise as the number of infected individuals reaches and exceeds a certain level. For example, the A/H1N1 virus was first reported in two US children in Mexico in March 2009, but health officials have stated that it apparently infected people as early as January 2009 in Mexico [9]. Hence, at the initial stage, the number of A/H1N1 cases and information about early precautionary measures against the disease were not available. With a diseases spreading further and information about infectious disease outbreaks being reported, media coverage about an epidemic disease can encourage the public to take precautionary measures against the disease such as wearing masks, avoiding public places, avoiding travel when sick, and frequent hand washing.

Therefore, it is necessary to refine the existing mathematical models with a media-induced effect to reflect this feature. Our main purpose is then to modify the known models in order to describe the impact of media coverage on disease transmission by introducing a piecewise continuous transmission term. We shall examine the effect of media coverage on disease transmission and to address whether the improved function representing a media-induced effect influences the global dynamics of disease transmission.

The rest of the paper is organized as follows. In Section 2, we propose a Filippov epidemic model to describe the media-induced effect on the transmission of infectious diseases. In Section 3, sliding mode dynamics and the existence of a pseudo-equilibrium are investigated. In Section 4, the bifurcation set and the sliding bifurcation of the proposed system are discussed. The global dynamics under variable threshold levels is examined in Section 5, and a discussion and biological conclusion are given in Section 6. Finally, for readers' interests, we present a brief introduction to threshold policy in the Appendix.

2. Filippov epidemic model and preliminaries

When the number of infected individuals increases and reaches a certain level I_c , mass media often start to report information about the disease, including ways of transmission, number of infected individuals, and then the public try their best to avoid being infected. This consequently lowers the effective contact, resulting in a reduction in transmission rate which is usually represented by $\exp(-\alpha I)$. We consider a population that is divided into three types: susceptible, infective, and recovered. Let S , I , and R denote the proportions of susceptible, infective, and recovered individuals with respect to the total population of individuals, respectively. Then the model equations are

$$\begin{cases} \frac{dS(t)}{dt} = \mu - \beta \exp(-\alpha \epsilon I) SI - \mu S, \\ \frac{dI(t)}{dt} = \beta \exp(-\alpha \epsilon I) SI - (\mu + \gamma) I, \\ \frac{dR(t)}{dt} = \gamma I - \mu R, \end{cases} \quad (1)$$

with

$$\epsilon = \begin{cases} 0, & \sigma(S, I) < 0, \\ 1, & \sigma(S, I) > 0, \end{cases} \quad (2)$$

where $\sigma(S, I)$ is a given function which may depend on the number of susceptible and infected individuals, and ϵ is a discontinuous control function. But here we choose the special function $\sigma(S, I) = \sigma(I) = I - I_c$, which means that only the number of infected individuals is the index for mass media to exhibit its effect. We note that model (1) with $\epsilon = 0$ was introduced and examined in [10]. All other parameters are positive constants, where μ is the natural birth (death) rate, β denotes the basic transmission rate, and γ represents the removed/recovered rate. Model (1) with (2) reveals a dynamical system subject to a threshold policy (TP), which is referred to as an on–off control or as a special and simple case of variable structure control in the control literature. A more detailed introduction on the TP is presented in the Appendix.

Since the recovered class R does not influence the dynamics of the first and second equations of model (1), we only need to focus on the first two equations in the rest of this work. It is easy to show that

$$\{(S, I) \in \mathbb{R}_+^2 \mid 0 < S(t) + I(t) \leq 1\} \doteq \Omega$$

is an attraction region of system (1). In fact, it follows from model (1) that $S(t) + I(t) + R(t) = 1$ and

$$\left. \frac{dS}{dt} \right|_{S=0} = \mu > 0, \quad \left. \frac{dI}{dt} \right|_{I=0} = 0, \quad \left. \frac{d(S+I)}{dt} \right|_{S+I=1} = \mu - \mu(S+I) - \gamma I < 0,$$

so all solutions will enter into the region Ω and remain in it ultimately. Hence, Ω is an attraction region. The (S, I) phase plane is split into three parts: $G^1 = \{(S, I) \in \mathbb{R}_+^2 \mid \sigma(I) < 0\}$, $G^2 = \{(S, I) \in \mathbb{R}_+^2 \mid \sigma(I) > 0\}$, and the discontinuous boundary

$$\Sigma = \{(S, I) \in \mathbb{R}_+^2 \mid \sigma(I) = 0\}. \quad (3)$$

Thus, in region G^1 , there is no effect from media coverage, and the transmission rate β remains unchanged, while in region G^2 , media coverage exhibits its effect, and hence the transmission rate declines to $\beta \exp(-\alpha I)$.

For convenience, let the vector $Z = (S, I)^T$, and denote

$$\begin{aligned} F_{G^1}(Z) &= (\mu - \beta SI - \mu S, \beta SI - (\mu + \gamma)I)^T, \\ F_{G^2}(Z) &= (\mu - \beta \exp(-\alpha I)SI - \mu S, \beta \exp(-\alpha I)SI - (\mu + \gamma)I)^T. \end{aligned}$$

Then system (1) with (2) can be rewritten as the following Filippov system [12]:

$$\dot{Z} = \begin{cases} F_{G^1}(Z), & Z \in G^1, \\ F_{G^2}(Z), & Z \in G^2. \end{cases} \quad (4)$$

The main characteristic of a Filippov system is that control is suppressed when the value of the threshold function is below a previously chosen threshold policy; above the threshold, control is applied, while in classical systems, control has been implemented all along [13–15]. In the following, we call Filippov system (4) defined in region G^1 system S^1 and that defined in region G^2 system S^2 .

Since the right-hand side of system (4) is piecewise continuous, we consider its solutions in Filippov's sense throughout the paper, which refers to solutions of the differential inclusions $\dot{Z}(t) \in F(Z(t))$, where

$$F(Z(t)) = \bigcap_{\delta > 0} \bigcap_{\mu N = 0} \overline{\text{co}}(F_{G^1}(B(Z; \delta) \setminus N) \cup F_{G^2}(B(Z; \delta) \setminus N)),$$

$\overline{\text{co}}$ stands for the convex hull, $B(Z; \delta)$ is the open δ -neighborhood of Z , and μ denotes the Lebesgue measure. It follows from the definition of F_{G^i} , $i = 1, 2$, that $F(Z(t))$ is nonempty, bounded and closed, convex, and upper semicontinuous in Z . Then, by [12], one gets that there exists a solution for system (4) with initial value $S(t_0) = S_0$, $I(t_0) = I_0$.

2.1. Dynamics for system S^1

We note that system S^1 has been studied in [10,11], but for clarity we present the main result and main method here. It is easy to obtain the basic reproduction number $R_0 = \beta/(\mu + \gamma)$.

Proposition 1. *The disease-free equilibrium $E_0 = (1, 0)$ is globally asymptotically stable if $R_0 < 1$, while the unique endemic equilibrium $E^1 = (S_*^1, I_*^1)$ is feasible and globally asymptotically stable if $R_0 > 1$, where $S_*^1 = 1/R_0$, $I_*^1 = \mu/\beta(R_0 - 1)$.*

Proof. If $R_0 < 1$, taking a Lyapunov function $V(t) = I(t)$ and applying LaSalle's invariance principle, one gets that E_0 is globally asymptotically stable. If $R_0 > 1$, we choose a Lyapunov function

$$V(t) = \frac{1}{2}\omega_1(S - S_*^1)^2 + \omega_1 S_*^1 \left(I - I_*^1 - I_*^1 \ln \frac{I}{I_*^1} \right).$$

Then, by using LaSalle's invariance principle, we conclude that the endemic equilibrium E^1 is globally asymptotically stable.

Moreover, the endemic equilibrium E^1 is a stable focus if $\Delta_1 < 0$, while it is a stable node if $\Delta_1 \geq 0$, where $\Delta_1 = \beta^2\mu - 4(\mu + \gamma)^2[\beta - (\mu + \gamma)]$.

2.2. Dynamics for system S^2

For system S^2 , the basic reproduction number is also R_0 , and the disease-free equilibrium is E_0 . The possible endemic state $E^2 = (S_*^2, I_*^2)$ satisfies

$$S_*^2 = \frac{\mu + \gamma}{\beta} \exp(\alpha I_*^2), \quad \mu - (\mu + \gamma)I_*^2 - \frac{\mu(\mu + \gamma)}{\beta} \exp(\alpha I_*^2) = 0.$$

Denote

$$g(I) \doteq \mu - (\mu + \gamma)I - \frac{\mu(\mu + \gamma)}{\beta} \exp(\alpha I);$$

then we have

$$g(0) = \mu - \frac{\mu(\mu + \gamma)}{\beta} > 0$$

provided that $R_0 > 1$ and

$$g\left(\frac{\mu}{\mu + \gamma}\right) = -\frac{\mu(\mu + \gamma)}{\beta} \exp\left(\frac{\alpha\mu}{\mu + \gamma}\right) < 0.$$

Further, simple calculation yields

$$g'(I) = -(\mu + \gamma) - \frac{\alpha\mu(\mu + \gamma)}{\beta} \exp(\alpha I) < 0$$

for all $I \geq 0$. All those indicate that there exists a unique I such that $g(I) = 0$, denoted by I_*^2 and such that $0 < I_*^2 < \mu/(\mu + \gamma)$, which can be determined analytically. In fact, $g(I_*^2) = 0$ is equivalent to

$$\frac{\alpha\mu}{\beta} \exp\left(\frac{\alpha\mu}{\mu + \gamma}\right) = \alpha\left(\frac{\mu}{\mu + \gamma} - I_*^2\right) \exp\left(\alpha\left(\frac{\mu}{\mu + \gamma} - I_*^2\right)\right).$$

According to the definition of the Lambert W function and solving the above equation with respect to I_*^2 , one obtains

$$I_*^2 = \frac{\mu}{\mu + \gamma} - \frac{1}{\alpha} \text{Lambert } W\left(\frac{\alpha\mu}{\beta} \exp\left(\frac{\alpha\mu}{\mu + \gamma}\right)\right).$$

Consequently, S_*^2 can be determined. The definition and properties of the Lambert W function can be found in the literature [16].

Proposition 2. *The disease-free equilibrium E_0 is globally asymptotically stable if $R_0 < 1$, while the endemic state E^2 is globally asymptotically stable if $R_0 > 1$.*

Proof. If $R_0 < 1$, we take a Lyapunov function $V(t) = I(t)$, and the same discussion as that in the proof of Proposition 1 yields that E_0 is globally asymptotically stable. If $R_0 > 1$, it is easy to get that the Jacobian matrix of S^2 at E^2 possesses two eigenvalues with negative real parts, so it is locally asymptotically stable. Further, letting a Dulac function $B = 1/(SI)$ and using Bendixson–Dulac criteria, we can exclude the existence of limit cycles, and hence E^2 is globally asymptotically stable.

Moreover, it is a stable focus if $\Delta_2 < 0$, where

$$\Delta_2 = \left(\frac{\mu}{S_*^2} + \alpha I_*^2(\mu + \gamma)\right)^2 - 4 \frac{I_*^2(\mu + \gamma)}{S_*^2} (\alpha\mu S_*^2 + \mu + \gamma).$$

Note that the equilibria E^1 and E^2 are endemic equilibria for subsystems S^1 and S^2 , respectively. However, they could be either in region G^1 or in G^2 . If the equilibrium points are located in their opposite regions, they are named virtual equilibrium points. Otherwise, they are called real equilibrium points. Both real equilibrium and virtual equilibrium are called standard equilibrium or regular equilibrium (see the definitions in [17,18] or the Appendix).

3. Sliding domain and its dynamics

There may exist sliding segments on the boundary (3) if there are some subregions of the line Σ such that both vectors of the two systems S^1 and S^2 are directed towards each other. Mathematically, there are two methods to determine sufficient conditions for a sliding mode to occur on the discontinuous surface: the well-known Filippov convex method [12] and Utkin's equivalent control method [18]. In the following, we employ the equivalent control method to determine the sliding domain and sliding mode dynamics of Filippov system (4).

Let

$$h(Z) = \langle \sigma_Z(Z), F_{G^1}(Z) \rangle \langle \sigma_Z(Z), F_{G^2}(Z) \rangle \doteq F_{G^1}\sigma(Z) \cdot F_{G^2}\sigma(Z),$$

where $\langle \cdot \rangle$ denotes the standard scalar product and $F\sigma(Z) = \langle \sigma_Z(Z), F(Z) \rangle$ is the Lie derivative of σ with respect to the vector field F at Z . Then the sliding domain can be defined as

$$\Sigma_S = \left\{ Z \in \Sigma \mid F_{G^1}\sigma(Z) \geq 0, F_{G^2}\sigma(Z) \leq 0 \right\}.$$

By the definition of the switching surface $\sigma(Z)$ (i.e., $\sigma(Z) = I - I_c$), and solving the two inequalities given in the above equation, one obtains the sliding domain of the Filippov system (4) as follows:

$$\Sigma_S = \left\{ (S, I) \in R_+^2 \mid \frac{\mu + \gamma}{\beta} \leq S \leq \frac{\mu + \gamma}{\beta} \exp(\alpha I_c), I = I_c \right\}. \quad (5)$$

Next, we address the dynamics of Filippov system (4) on the boundary Σ_S [18]. Note that $\sigma' = I' = \beta \exp(-\alpha\epsilon I)SI - (\mu + \gamma)I$. Let $\sigma' = 0$; solving with respect to ϵ , we have

$$\epsilon = -\frac{1}{\alpha I} \ln\left(\frac{\mu + \gamma}{\beta S}\right).$$

Substituting this and $\sigma = I - I_c = 0$ into the first equation of model (1), we obtain the sliding mode dynamics, which can be determined by the following differential equation:

$$S' = \mu - \mu S - (\mu + \gamma)I_c \quad (6)$$

with

$$\frac{\mu + \gamma}{\beta} \leq S \leq \frac{\mu + \gamma}{\beta} \exp(\alpha I_c).$$

It follows that there is an equilibrium $S_* = (\mu - (\mu + \gamma)I_c)/\mu$ for system (6) provided that

$$\frac{\mu + \gamma}{\beta} < S_* = \frac{\mu - (\mu + \gamma)I_c}{\mu} < \frac{\mu + \gamma}{\beta} \exp(\alpha I_c),$$

which is equivalent to

$$I_*^2 < I_c < I_*^1, \quad (7)$$

where I_*^1 and I_*^2 are defined in Section 2.

Since the components of both E^1 and E^2 satisfy $\mu - \mu S - (\mu + \gamma)I = 0$ and $S_*^1 < S_*^2$, we know that $I_*^1 > I_*^2$, which implies that the inequalities (7) are well defined. Note that the equilibrium $E_S = (S_*, I_*) = ((\mu - (\mu + \gamma)I_c)/\mu, I_c)$ is called a pseudo-equilibrium of Filippov system (4) according to Definition A.2 in the Appendix.

In order to analyze the relation between the sliding segment Σ_S and the attraction region Ω , we need to solve the following equations:

$$\frac{\mu + \gamma}{\beta} = 1 - I \quad \text{and} \quad \frac{\mu + \gamma}{\beta} \exp(\alpha I) = 1 - I.$$

It is easy to derive

$$I_c^1 = \frac{\beta - (\mu + \gamma)}{\beta}$$

from the first equation. For the second equation, denote $h(I) = (\mu + \gamma)/\beta \exp(\alpha I) + I - 1$, and by implementing the similar discussion for the function $g(I)$, one can get that there exists a unique I satisfying $h(I) = 0$ and $0 < I < 1$, denoted by I_c^2 , which can be solved explicitly. In fact, $h(I_c^2) = 0$ can be written as

$$\frac{\alpha(\mu + \gamma)}{\beta} \exp(\alpha) = \alpha(1 - I_c^2) \exp(\alpha(1 - I_c^2)),$$

solving which yields

$$I_c^2 = 1 - \frac{1}{\alpha} \text{Lambert } W \left(\frac{\alpha(\mu + \gamma)}{\beta} \exp(\alpha) \right).$$

Then we can get that the sliding segment Σ_S is totally out of the attraction region Ω if $I_c > I_c^1$, while it lies totally in Ω for $I_c < I_c^2$. If $I_c^2 < I_c < I_c^1$, part of Σ_S is out of the attraction region Ω and part of it is in Ω .

4. Sliding bifurcation analysis

Bifurcation sets of equilibria and sliding modes are related to the sliding bifurcations, and all these are useful for the forthcoming global dynamics analysis.

4.1. Bifurcation sets of equilibria and sliding modes

In this subsection, we will address the richness of the possible equilibria and sliding modes that the system can exhibit. Parameters α and I_c are chosen to build the bifurcation diagram, and all other parameters are fixed as those in Fig. 1. We define the four curves for parameters α and I_c as follows: $L_1 = \{(\alpha, I_c) | I_c = I_*^1\}$, $L_2 = \{(\alpha, I_c) | I_c = I_*^2\}$, $L_3 = \{(\alpha, I_c) | I_c = I_c^1\}$, and $L_4 = \{(\alpha, I_c) | I_c = I_c^2\}$, where I_*^i , I_c^i , $i = 1, 2$, are defined as in previous sections.

The two solid lines L_1 and L_2 divide the $\alpha - I_c$ parameter space into three parts, and the coexistence of regular, virtual, or pseudo equilibria are indicated in each part. When $I_c > I_*^1$ (i.e., the region $\Gamma_1 \cup \Gamma_2 \cup \Gamma_3$), the equilibrium E^1 is real, while E^2 is virtual (denoted by E_r^1 and E_v^2 , respectively), and E_S does not exist. When $I_*^2 < I_c < I_*^1$ (i.e., the region $\Gamma_4 \cup \Gamma_5$), both the equilibria E^1 and E^2 are virtual (denoted by E_v^1 and E_v^2 , respectively), and E_S does exist. When $I_c < I_*^2$ (i.e., the region Γ_6), the equilibrium E^2 is real, while E^1 is virtual (denoted by E_v^1 and E_r^2 , respectively), and E_S does not exist, either. It should be

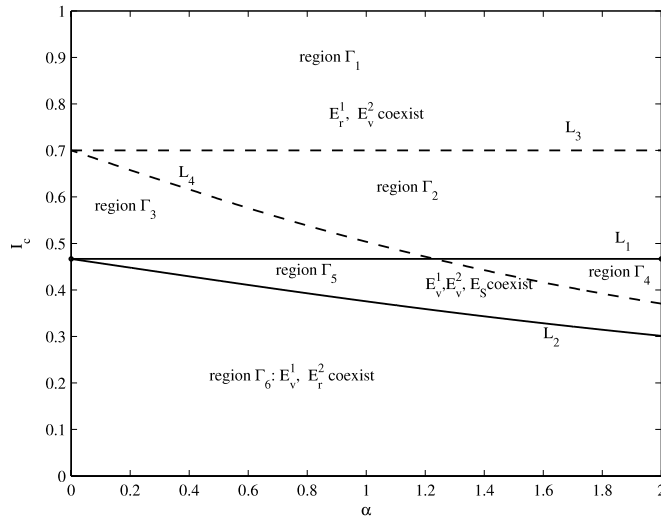


Fig. 1. Bifurcation set for Filippov system (4) with respect to α and I_c . Here we fixed all other parameters as follows: $\mu = 0.2$; $\beta = 1$; $\gamma = 0.1$.

noted that the equilibria of system (4) actually include the equilibria of S^1 or S^2 , which may be real or virtual equilibria, and pseudo-equilibria, if feasible.

The two dashed lines L_3 and L_4 also divide the parameter space into three parts. In region Γ_1 , the sliding segment Σ_S is totally out of the attraction region Ω . In region $\Gamma_2 \cup \Gamma_4$, Σ_S is partly out of the attraction region Ω and it lies totally in Ω in region $\Gamma_3 \cup \Gamma_5 \cup \Gamma_6$.

This implies that, as the threshold level I_c increases, the sliding segment Σ_S will enlarge and may intersect with the boundary of the attraction region Ω such that part of it is in the attraction region or even lies out of the attraction region ultimately.

It is clear that, if one or more parameters changes, some bifurcations occur in Fig. 1. For example, boundary focus bifurcation as well as boundary node bifurcation can occur when one of the parameters varies such as I_c , which will be addressed in detail in the following subsection.

4.2. Boundary focus bifurcation

In this subsection, we fix all other parameters and choose the threshold value I_c as the bifurcation parameter. By Section 2, we know that E^i is a focus (node) if $\Delta_i < 0$ ($\Delta_i \geq 0$), where $i = 1, 2$. In the following, we only investigate the boundary focus bifurcation (i.e., $\Delta_i < 0$); the boundary node bifurcation (i.e., $\Delta_i \geq 0$) can be discussed similarly. To clearly show the boundary focus bifurcation of Filippov system (4), we first discuss the following two types of special point named the tangent point (denoted as T) and the boundary equilibrium (denoted as E_b). See [19,20] and the Appendix for detailed definitions of the tangent point and boundary equilibrium.

Tangent points of Filippov system (4) satisfy

$$\beta \exp(-\alpha \epsilon I) S I - (\mu + \gamma) I = 0, \quad I = I_c,$$

solving which with respect to S yields the following two tangent points:

$$T_1 = \left(\frac{\mu + \gamma}{\beta}, I_c \right) \quad \text{and} \quad T_2 = \left(\frac{\mu + \gamma}{\beta} \exp(\alpha I_c), I_c \right).$$

It is clear that points T_1 and T_2 are the endpoints of the sliding segment Σ_S .

The boundary equilibrium of Filippov system (4) satisfies

$$\mu - \beta \exp(-\alpha \epsilon I) S I - \mu S = 0, \quad \beta \exp(-\alpha \epsilon I) S I - (\mu + \gamma) I = 0, \quad I = I_c.$$

We require the equalities

$$\frac{\mu + \gamma}{\beta \exp(-\alpha \epsilon I_c)} = \frac{\mu}{\mu + \beta \exp(-\alpha \epsilon I_c) I_c}$$

to hold true to ensure the existence of boundary equilibria, which indicates that $I_c = I_*^1$ for $\epsilon = 0$, while, for $\epsilon = 1$, it is equivalent to

$$\frac{\alpha \mu}{\beta} \exp\left(\frac{\alpha \mu}{\mu + \gamma}\right) = \left(\frac{\alpha \mu}{\mu + \gamma} - \alpha I_c\right) \exp\left(\frac{\alpha \mu}{\mu + \gamma} - \alpha I_c\right),$$

solving which gives $I_c = I_*^2$. Hence, there exists boundary equilibrium

$$E_b^1 = \left(\frac{\mu}{\mu + \beta I_c}, I_c \right) \quad \text{or} \quad E_b^2 = \left(\frac{\mu}{\mu + \beta \exp(-\alpha I_c) I_c}, I_c \right)$$

provided that $I_c = I_*^1$ or $I_c = I_*^2$.

If the pseudo-equilibrium, tangent point, and real equilibrium (or tangent point and real equilibrium) collide simultaneously when the parameter I_c passes through a critical value, boundary focus bifurcation may occur. For example, when parameter I_c passes through the first critical value $I_c = I_*^1$, the equilibrium E_r^1 and tangent point T_1 can collide, as shown in Fig. 2(b), where $I_*^1 = 0.4667$. In this case, the boundary equilibrium E_b^1 is an attractor (see Fig. 2(b)). A stable focus E_r^1 and a tangent point T_1 coexist for $I_c > I_*^1$, as shown in Fig. 2(a) with $I_c = 0.5$. They collide at $I_c = 0.4667$ and are replaced by a pseudo-equilibrium E_s and a tangent point T_1 for $I_c < I_*^1$, as shown in Fig. 2(c) with $I_c = 0.43$. This bifurcation shows how a stable focus can become a stable pseudo-equilibrium.

Another boundary focus bifurcation of Filippov system (4) occurs when parameter I_c passes through the second critical value $I_c = I_*^2$. In this case, a stable focus E_r^2 and a tangent point T_2 coexist for $I_c < I_*^2$. They collide at $I_c = I_*^2$ and are replaced by a pseudo-equilibrium E_s and a tangent point T_2 for $I_c > I_*^2$.

5. Global behavior of the system

We shall investigate the global stability of all the possible equilibria, including the standard equilibrium and pseudo-equilibrium, under different threshold levels and $R_0 > 1$. If $R_0 < 1$, the disease is eradicated, and it is not necessary to consider any media-induced effect; thus we only focus on the case when $R_0 > 1$.

5.1. Nonexistence of limit cycles

Indeed, to prove the global stability of the equilibrium of system (4), we need to rule out the existence of limit cycles. First, we shall preclude the existence of limit cycles totally in region G^1 or G^2 . Denote the right-hand sides of system S^i by $f^{(i)}(x)$, where $f^{(i)}(x) = (f_1^{(i)}(x), f_2^{(i)}(x))$, $i = 1, 2$. Let the Dulac function be $B = 1/(SI)$. For system S^1 , we have

$$\frac{\partial(Bf_1^{(1)})}{\partial S} + \frac{\partial(Bf_2^{(1)})}{\partial I} = -\frac{\mu}{S^2 I} < 0,$$

so there is no limit cycle totally in G^1 . Similarly, for system S^2 , we get

$$\frac{\partial(Bf_1^{(2)})}{\partial S} + \frac{\partial(Bf_2^{(2)})}{\partial I} = -\frac{\mu}{S^2 I} - \alpha \beta \exp(-\alpha I) < 0,$$

so no limit cycle is totally in G^2 .

By the above discussion, we easily obtain the following result.

Lemma 1. *There is no limit cycle for Filippov system (4) which is totally in region G^i ($i = 1, 2$).*

Next, we shall preclude the existence of limit cycles, pieces of which are in the sliding region Σ_S .

Lemma 2. *There is no closed orbit for Filippov system (4) which contains part of the sliding segment $T_1 T_2$.*

Proof. By Section 4.1, there does exist a pseudo-equilibrium E_s for $I_*^2 < I_c < I_*^1$ which is asymptotically stable in the sliding domain Σ_S . The local stability of the pseudo-equilibrium E_s on the sliding domain indicates that no limit cycle containing part of the sliding segment exists in such a case. Therefore, we only need to show the result for $I_c > I_*^1$ or for $I_c < I_*^2$. In the following, we shall do it for $I_c > I_*^1$; for $I_c < I_*^2$, we can use a similar process.

In fact, since $I_c > I_*^1$, by the expression of the sliding domain given in (5), the sliding mode dynamics (6) satisfies $S' < 0$, which shows that the trajectory moves from the right to the left on the sliding segment $T_1 T_2$ (as shown in Fig. 3).

We only need to show that the orbit starting from the tangent point T_1 , denoted by l_1 , cannot hit the manifold Σ again. Since E_r^1 is a stable focus for $\Delta_1 < 0$, the orbit l_1 intersects with the horizontal isocline g_1^1 , where g_1^1 refers to $\{(S, I) | S = (\mu + \gamma)/\beta, I < I_c\}$, at point M_1 first, and at point N_1 second. Hence we conclude that point N_1 lies between T_1 and E_r^1 . Otherwise, the trajectory l_1 may either coincide with point T_1 or hit the manifold Σ at some point to the right of point T_1 . If the former is true, l_1 is a closed orbit which is tangent to the sliding segment $T_1 T_2$. Then any trajectory starting from a point out of the closed orbit l_1 cannot cross the cycle l_1 , and certainly cannot tend to the endemic equilibrium E_r^1 , which contradicts the stability of equilibrium E_r^1 in G^1 . If the latter holds, it indicates that the endemic state E_r^1 is an unstable focus, which also contradicts that E_r^1 is stable in G^1 . Hence the orbit initiating from point T_1 cannot hit the manifold Σ but spirally approaches the stable equilibrium E_r^1 . Therefore, no closed orbit containing part of the switching segment $T_1 T_2$ exists. This completes the proof.

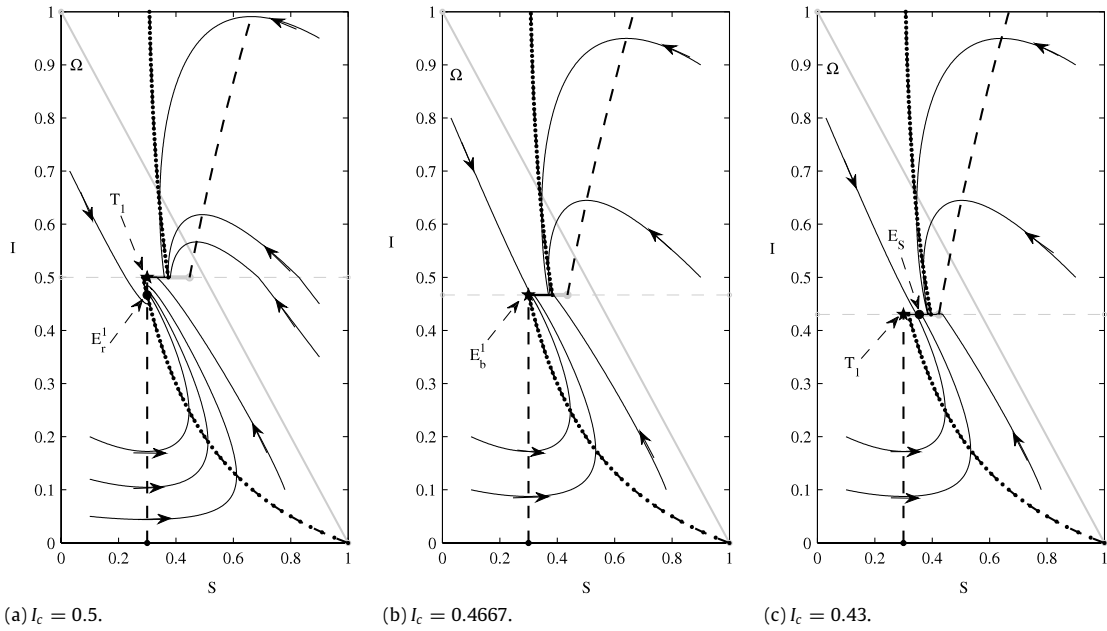


Fig. 2. Boundary focus bifurcation for Filippov system (4). Here we choose I_c as a bifurcation parameter and fix all other parameters as follows: $\mu = 0.2$, $\beta = 1$, $\gamma = 0.1$, $\alpha = 0.8$, $I_c = 0.5$ (a), $I_c = 0.4667$ (b), $I_c = 0.43$ (c).

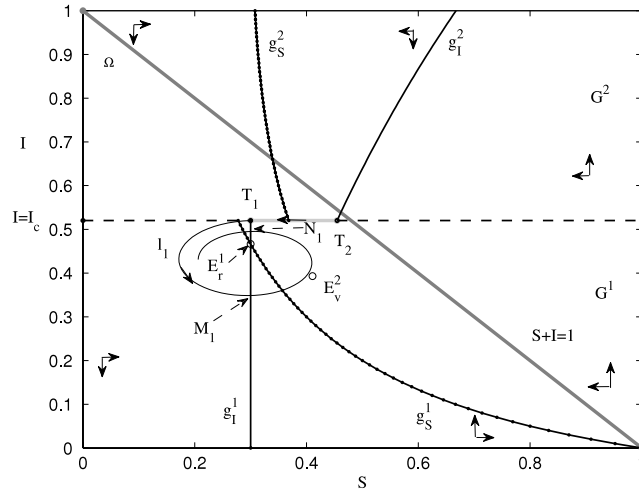


Fig. 3. Phase plane S – I for Filippov system (4), showing the switching line ($I = I_c$), sliding domain (T_1T_2), real equilibrium (small circle: E_r^1), virtual equilibrium (small circle: E_b^1), and the attraction region Ω bounded by the line $S + I = 1$. The horizontal and vertical isoclinic curves for the free system S^1 are $\{(S, I) \in R_+^2 | S = (\mu + \gamma)/\beta, I < I_c\}$ and $\{(S, I) \in R_+^2 | I = -\mu/\beta + \mu/(\beta S), I < I_c\}$, denoted by g_I^1 and g_S^1 , respectively. The horizontal and vertical isoclinic curves for the controlled system S^2 are $\{(S, I) \in R_+^2 | I = 1/\alpha \ln(\beta S/(\mu + \gamma)), I > I_c\}$ and $\{(S, I) \in R_+^2 | I = -1/\alpha \text{Lamber } W(\alpha\mu/\beta - \alpha\mu/(\beta S)), I > I_c\}$, denoted by g_I^2 and g_S^2 , respectively. The typical orbit I_1 is plotted simply to show the asymptotical stability of the focus E_r^1 for Case 1 without great accuracy.

We now rule out the existence of the limit cycle induced by the nonsmooth nature of the trajectories on the complement to Σ_S in Σ ; that is, we will prove that there is no limit cycle surrounding the sliding segment T_1T_2 which is partly in region G^1 and partly in region G^2 . So, we give the following result when the sliding segment is totally within the attraction region.

Lemma 3. *There is no limit cycle surrounding the sliding segment T_1T_2 in the attraction region Ω .*

Proof. The method of proving this lemma is similar to that in [21,22], but for readers interested in the technical aspects we give the details in the following. Suppose that X is a limit cycle in the interior of the attraction region Ω , and that it surrounds the sliding segment T_1T_2 with a period T , as shown in Fig. 4. Denote its part below the line $I = I_c$ by X_1 and its part above the line $I = I_c$ by X_2 . Denote the intersection points of the limit cycle X and the line $I = I_c$ by H_1 and H_2 , the intersection points of X and the auxiliary line $I = I_c - \epsilon$ by A_1 and A_2 , and the intersection points of X and another auxiliary

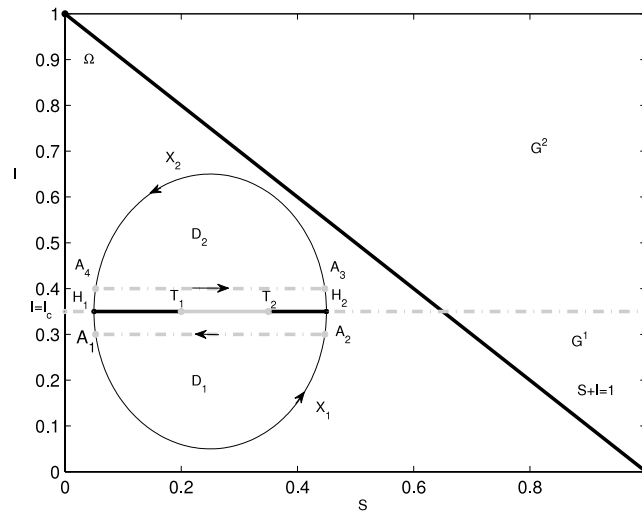


Fig. 4. Phase plane S – I for Filippov epidemic model (4), showing the switching line ($I = I_c$), sliding domain (T_1T_2), and the diagram of X_1 and X_2 split from the limit cycle by the switching line.

line $I = I_c + \epsilon$ by A_4 and A_3 , where ϵ ($\epsilon > 0$) is any sufficiently small number. Let D_1 be the region delimited by X_1 and the segment A_2A_1 , and let D_2 be the region delimited by X_2 and the segment A_4A_3 . We denote the boundary of D_1 and D_2 by C_1 and C_2 , respectively, with the directions indicated in the Fig. 4. Let the Dulac function be $B = 1/(SI)$ defined as before. By the discussion given above and Green's theorem, we get

$$\begin{aligned} \iint_{D_1} \left[\frac{\partial(Bf_1^{(1)})}{\partial S} + \frac{\partial(Bf_2^{(1)})}{\partial I} \right] dS dI &= \oint_{C_1} B[f_1^{(1)} dI - f_2^{(1)} dS] = - \int_{A_2A_1} Bf_2^{(1)} dS, \\ \iint_{D_2} \left[\frac{\partial(Bf_1^{(2)})}{\partial S} + \frac{\partial(Bf_2^{(2)})}{\partial I} \right] dS dI &= \oint_{C_2} B[f_1^{(2)} dI - f_2^{(2)} dS] = - \int_{A_4A_3} Bf_2^{(2)} dS. \end{aligned}$$

Let $D_0 \subset D_1$ and

$$\xi = \iint_{D_0} \left[\frac{\partial(Bf_1^{(1)})}{\partial S} + \frac{\partial(Bf_2^{(1)})}{\partial I} \right] dS dI;$$

then, since we know that $\xi < 0$, and, furthermore,

$$0 > \xi > \iint_{D_1} \left[\frac{\partial(Bf_1^{(1)})}{\partial S} + \frac{\partial(Bf_2^{(1)})}{\partial I} \right] dS dI + \iint_{D_2} \left[\frac{\partial(Bf_1^{(2)})}{\partial S} + \frac{\partial(Bf_2^{(2)})}{\partial I} \right] dS dI,$$

we obtain

$$0 > \xi > - \int_{A_2A_1} Bf_2^{(1)} dS - \int_{A_4A_3} Bf_2^{(2)} dS. \quad (8)$$

Suppose that the abscissas of the points $H_1, H_2, A_1, A_2, A_4, A_3$ are $\underline{S}, \bar{S}, \underline{S} + h_1(\epsilon), \bar{S} - h_2(\epsilon), \underline{S} + h_4(\epsilon), \bar{S} - h_3(\epsilon)$, respectively, where $h_i(\epsilon)$ ($i = 1, 2, 3, 4$) is continuous and satisfies $\lim_{\epsilon \rightarrow 0} h_i(\epsilon) = 0$ and $h_i(0) = 0$. Thus, we get

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \left[- \int_{A_2A_1} Bf_2^{(1)} dS \right] &= \lim_{\epsilon \rightarrow 0} \int_{\underline{S}+h_1(\epsilon)}^{\bar{S}-h_2(\epsilon)} \left(\beta - \frac{\mu + \gamma}{S} \right) dS \\ &= \lim_{\epsilon \rightarrow 0} \left[\beta(\bar{S} - \underline{S} - h_2(\epsilon) - h_1(\epsilon)) - (\mu + \gamma) \ln \left| \frac{\bar{S} - h_2(\epsilon)}{\underline{S} + h_1(\epsilon)} \right| \right] \\ &= \beta(\bar{S} - \underline{S}) - (\mu + \gamma) \ln \left| \frac{\bar{S}}{\underline{S}} \right|. \end{aligned}$$

By the same process, we get

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \left[- \int_{A_4 A_3} B f_2^{(2)} dS \right] &= - \lim_{\epsilon \rightarrow 0} \int_{\underline{S} + h_4(\epsilon)}^{\bar{S} - h_3(\epsilon)} \left(\beta \exp(-\alpha(I_c + \epsilon)) - \frac{\mu + \gamma}{S} \right) dS \\ &= - \lim_{\epsilon \rightarrow 0} \left[\beta \exp(-\alpha(I_c + \epsilon)) (\bar{S} - \underline{S} - h_3(\epsilon) - h_4(\epsilon)) - (\mu + \gamma) \ln \left| \frac{\bar{S} - h_3(\epsilon)}{\underline{S} + h_4(\epsilon)} \right| \right] \\ &= -\beta \exp(-\alpha I_c) (\bar{S} - \underline{S}) + (\mu + \gamma) \ln \left| \frac{\bar{S}}{\underline{S}} \right|. \end{aligned}$$

Therefore,

$$\lim_{\epsilon \rightarrow 0} \left[- \int_{A_2 A_1} B f_2^{(1)} dS - \int_{A_4 A_3} B f_2^{(2)} dS \right] = \beta(1 - \exp(-\alpha I_c))(\bar{S} - \underline{S}) > 0,$$

which contradicts (8). This rules out the existence of the limit cycle X . This completes the proof.

5.2. Global stability of the equilibria

To establish all the possible behaviors that the system can exhibit, we choose variable sets of values for the parameters such that the dynamics in all regions are presented. In this subsection, we consider the following three cases.

Case 1. Suppose that $I_c > I_*^1$. Then equilibrium E^1 is real, while E^2 is virtual. By calculation, we know that the sliding mode does exist, but there is not a pseudo-equilibrium. In the following, we shall show that E_r^1 is globally asymptotically stable.

Theorem 1. For Filippov system (4), the endemic equilibrium E_r^1 is globally asymptotically stable if $R_0 > 1$.

Proof. Suppose that $\Delta_1 < 0$. Then the endemic equilibrium E_r^1 is a focus, which is locally asymptotically stable, by the above discussion. The procedure of the proof for Lemma 2 shows that the trajectory moves from the right to the left on the sliding segment $T_1 T_2$ in this case.

Note that it is easy to get $I_c^1 > I_*^1$, $I_c^2 > I_*^2$ and $I_c^1 \geq I_c^2$, where I_c^1 and I_c^2 are defined as those in Section 3. Then, either $I_c^2 > I_*^1$ or $I_c^2 < I_*^1$ may hold true. For different sets of values of the parameters, the sliding segment may exclusively, partly, or totally be in the attraction region Ω , so we then consider the following three possibilities.

(i) If $I_c \geq I_c^1$, then the point $(\alpha, I_c) \in \Gamma_1$ in the $\alpha - I_c$ plane (shown in Fig. 1). The sliding segment $T_1 T_2$ lies out of the attraction region Ω entirely, as shown in Fig. 5(a).

Then the regular equilibrium E_r^1 and the virtual equilibrium E_v^2 are in region $G^1 \cap \Omega$, and the direction of the vector field in region $G^2 \cap \Omega$ points downward. Hence, no limit cycle which is partly in region G^1 and partly in G^2 exists in the attraction region Ω . In addition, by Lemma 1, there is no limit cycle totally in G^i ($i = 1, 2$). Therefore the local stability of the endemic equilibrium E_r^1 implies that E_r^1 is globally asymptotically stable.

(ii) If $\max\{I_*^1, I_c^2\} < I_c < I_c^1$, then we get $(\alpha, I_c) \in \Gamma_2$ in the $\alpha - I_c$ plane (see Fig. 1). In such a case the sliding segment $T_1 T_2$ partly lies out of the attraction region Ω , as shown in Fig. 5(b). Note that again the direction of the vector field in region $G^2 \cap \Omega$ points downward. Moreover, according to Lemma 2, we know that no closed orbit containing part of the sliding segment $T_1 T_2$ exists. Hence, the endemic equilibrium E_r^1 is globally asymptotically stable.

(iii) If $I_c \leq \max\{I_*^1, I_c^2\}$, then we can get $(\alpha, I_c) \in \Gamma_3$ in the $\alpha - I_c$ plane (see Fig. 1), and the sliding segment is totally in the attraction region Ω , as shown in Fig. 5(c).

Note that a limit cycle totally in the region G^1 or G^2 does not exist. In addition, Lemma 2 precludes the possibility of a limit cycle induced by the sliding nature of the trajectories on the sliding domain Σ_S . It follows from Lemma 3 that there is no limit cycle in the attraction region Ω which surrounds the sliding segment $T_1 T_2$. Hence, the endemic equilibrium E_r^1 is globally asymptotically stable.

If $\Delta_1 \geq 0$, the endemic equilibrium E_r^1 is a node, which is also asymptotically stable in G^1 . Note that both the regular equilibrium E_r^1 and virtual equilibrium E_v^2 are in region G^1 . This indicates that all trajectories will attain the subregion of the attraction region Ω below the manifold Σ . Moreover, they will remain in it and approach the endemic equilibrium E_r^1 . Hence E_r^1 is globally asymptotically stable. This completes the proof of Theorem 1.

Case 2. If $I_*^2 < I_c < I_*^1$, both E^1 and E^2 are virtual. The sliding mode does exist, and there does exist a pseudo-equilibrium E_S which is asymptotically stable in the sliding domain $T_1 T_2$. We shall show that the pseudo-equilibrium E_S is globally asymptotically stable in the SI phase plane in the following.

Theorem 2. For Filippov system (4), the pseudo-equilibrium E_S is globally asymptotically stable if $R_0 > 1$.

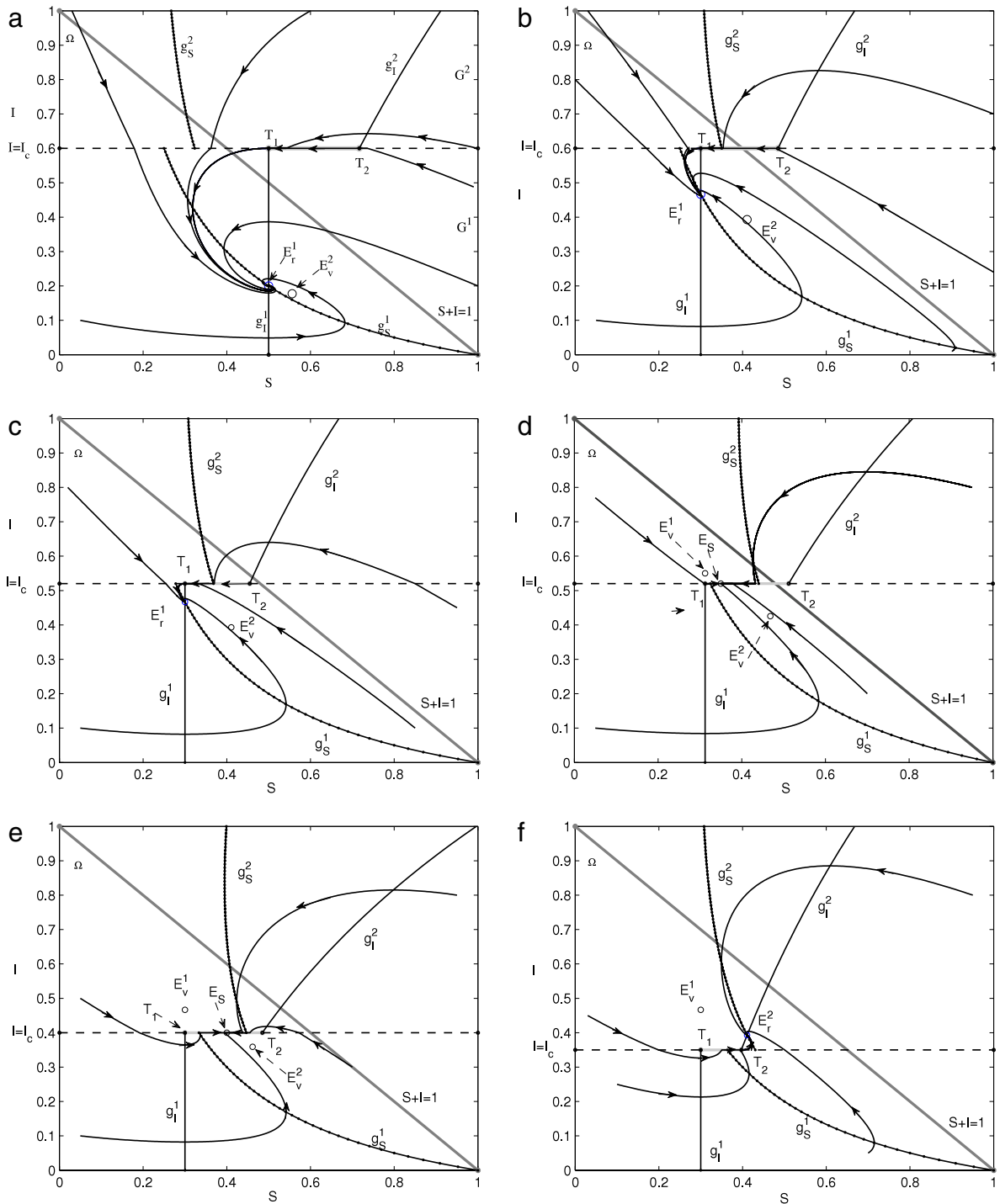


Fig. 5. Phase plane $S-I$ for Filippov epidemic model (4), showing the sliding domain (T_1T_2) and asymptotical equilibrium (small circles: E_r^1, E_r^2 or E_s^1, E_s^2 , $i = 1, 2$) for different parameter sets. The isoclinic lines g_1^1 (g_1^2) and g_2^1 (g_2^2) are plotted for the free (controlled) system S^1 (S^2). The curves represent the orbits in the phase plane indicating the asymptotical equilibrium. Parameters values are chosen such that dynamics in all region Γ_i , $i = 1, 2, \dots, 6$ as shown in Fig. 1 are exhibited. (a) $\mu = 0.2, \beta = 1, \gamma = 0.3, \alpha = 0.6, I_c = 0.6$ for region Γ_1 ; (b) $\mu = 0.2, \beta = 1, \gamma = 0.1, \alpha = 0.8, I_c = 0.6$ for region Γ_2 ; (c) $\mu = 0.2, \beta = 1, \gamma = 0.1, \alpha = 0.8, I_c = 0.52$ for region Γ_3 ; (d) $\mu = 0.2, \beta = 0.8, \gamma = 0.05, \alpha = 0.95, I_c = 0.52$ for region Γ_4 ; (e) $\mu = 0.1, \beta = 1, \gamma = 0.1, \alpha = 1.2, I_c = 0.4$ for region Γ_5 ; (f) $\mu = 0.2, \beta = 1, \gamma = 0.1, \alpha = 0.8, I_c = 0.35$ for region Γ_6 .

Proof. By Lemmas 1 and 2, we know that there is no limit cycle totally in the region G^i ($i = 1, 2$), or containing part of the sliding segment T_1T_2 . Note that the sliding segment may be partly or totally in the interior of the attraction region. Then there are two possibilities to consider.

(i) If $I_c \geq \min\{I_c^2, I_*^1\}$, it is easy to get $(\alpha, I_c) \in \Gamma_4$ in the $\alpha - I_c$ plane (see Fig. 1), and part of the sliding segment T_1T_2 is in the attraction region, as shown in Fig. 5(d). Note that again the direction of the vector field in region $G^2 \cap \Omega$ points downward. By using similar arguments as the second possibility of Case 1, we obtain that no limit cycle exists in the attraction region.

(ii) If $I_c < \min\{I_c^2, I_*^1\}$, then $(\alpha, I_c) \in \Gamma_5$ in the $\alpha - I_c$ plane (see Fig. 1), and the sliding segment is totally in the attraction region Ω , as shown in Fig. 5(e). Based on a similar argument to Lemma 3, we can exclude the existence of limit cycles in the attraction region Ω which surround the sliding segment T_1T_2 .

Note that both locally stable equilibria E^1 and E^2 are virtual and then cannot be attained. Consequently, any orbit initiating from region G^1 (G^2) intersects with the switching segment in order to approach the equilibrium point E_v^1 (E_v^2). Further, according to the above discussion, no limit cycle exists in the attraction region. We then conclude the pseudo-equilibrium E_S is globally asymptotically stable. This completes the proof of Theorem 2.

Case 3. If $I_c < I_*^2$, then we have $(\alpha, I_c) \in \Gamma_6$ in the $\alpha - I_c$ plane (see Fig. 1). The endemic equilibrium E^1 is virtual and E^2 is a real equilibrium, and there is no pseudo-equilibrium. The sliding mode does exist, and the trajectory moves from the left to the right on the sliding segment in such a case, as shown in Fig. 5(f). By a similar discussion to that in the third possibility of Case 1, we know that there is no limit cycle totally in the attraction region Ω . If the endemic equilibrium E_r^1 is locally stable, E_r^2 is globally asymptotically stable, and we get the following conclusion.

Theorem 3. For Filippov system (4), the endemic equilibrium E_r^2 is globally asymptotically stable if $R_0 > 1$.

Remark. If we consider the continuous transmission rate (i.e., $I_c = 0$), then system (1)–(2) becomes system S^2 . This indicates that the media coverage exhibits its effect once the epidemic disease emerges and begins to spread. By Proposition 2, we know that system S^2 will stabilize at the disease-free equilibrium E_0 or endemic equilibrium E^2 depending on the basic reproduction number R_0 . In fact, when $R_0 < 1$, the disease-free equilibrium E_0 is globally asymptotically stable, suggesting that the disease is eradicated; when $R_0 > 1$, the endemic equilibrium E^2 turns out to be a global attractor, suggesting that the disease persists and the number of infected individuals approaches the level of I_*^2 , which is determined uniquely by system S^2 itself.

However, Filippov system (4) is an epidemic system with piece-wise continuous transmission rate. It follows from Theorems 1 and 3 that it may stabilize at one of the endemic equilibria (i.e., E^1 or E^2) of the variable structures when $R_0 > 1$, depending on the choice of the critical level I_c . The most interesting and crucial issue is that the switched system (4) can stabilize at the pseudo-equilibrium E_S (Theorem 2), which is new and qualitatively different from the equilibrium for system S^2 . This indicates that a previously chosen level of the desired number of the infected persons can be reached when the threshold level and other parameters are chosen properly, which may suggest to us an idea for disease control when the emergent infectious diseases cannot be eradicated immediately.

6. Discussion and biological conclusion

A few mathematical models have described the impact of media coverage on the transmission dynamics of infectious diseases (see [1,3,5,6]). The formulated models implicitly assume that the media-induced effect occurs throughout all the epidemic time. However, public mass media usually cannot be aware of or report infectious diseases when the number of infected individuals is generally small at the initial stages of disease spreading. To model this vital fact, we extend the existing models by incorporating a piecewise continuous transmission rate to represent that media reports and education can exhibit their effects once the number of infected individuals exceeds a certain level. To our best knowledge, system (1) with (2) is the first epidemic system that incorporates a piecewise continuous transmission rate to describe the real impact of media coverage on infectious diseases.

The proposed epidemic model in this paper, which is described by a piecewise smooth ordinary differential equations, describes the media-induced effect subject to variable structure control. The system is actually called a Filippov system (or switching system) in control language [12]. A commonly used and implementable control is the simple on–off control, which can stabilize the models [21,23–28]. The overall objective of this paper is to examine the effect of media coverage on disease transmission and to address whether the improved function representing the media-induced effect influences the global dynamics of disease transmission.

The analysis performed in this work establishes all possible dynamic behaviors that the present model can exhibit, as can be seen in Figs. 2 and 5. We employ the properties of the Lambert W function [16] and bifurcation analysis of piecewise smooth systems [19,20] to investigate the long-term dynamical behavior of the proposed Filippov epidemic model. In particular, the sliding mode dynamics and the local sliding bifurcations have been addressed in this paper. The crucial role of the variable structure control in determining the dynamical behavior of the model is shown. We have also obtained the global stability for the Filippov epidemic model by using qualitative techniques. Our main results suggest that the system stabilizes at the equilibrium point E_r^1 , E_r^2 , or E_S , depending on the threshold level I_c . Six different types of threshold level are considered (see Fig. 5). For each one, the sliding segment is exclusively, partly, or totally in the attraction region. Fig. 5(a)–(c) show that the endemic state E_r^1 is globally asymptotically stable, i.e., E_r^1 is a global attractor. Trajectories may approach the endemic equilibrium E_r^1 without sliding, or enter the sliding region and then slide and approach the equilibrium E_r^1 , depending on the initial data. Similarly, Fig. 5(f) shows that the endemic equilibrium E_r^2 is an attractor.

The most interesting case is represented in Fig. 5(d) and (e), where the threshold level I_c is in between I_*^1 and I_*^2 (i.e., $I_*^1 < I_c < I_*^2$) and both the equilibria E^1 and E^2 are virtual, from which we know that the pseudo-equilibrium E_5 is globally asymptotically stable. Comparing with the former cases, trajectories will enter the sliding domain and slide to the pseudo-equilibrium E_5 . This implies that the infected population can stabilize at a previously chosen level (i.e., I_c) when the threshold policy and other parameters are chosen properly. Therefore, the threshold policy combines the characteristics of the free and controlled systems, and, in particular, induces a new equilibrium E_5 at which system (4) stabilizes compared to the dynamic behavior for system S^2 with continuous transmission rate incorporating media coverage (see [1,3,6]).

Although the threshold policy can significantly modify the dynamical behavior of the models studied, it is important to note that the number of infected individuals in the steady state triggered by the threshold policy is confined to a set of values determined previously by the parameters of the model, if some conditions on the parameters hold true. Hence, the policy is robust [17] to bounded uncertainty in the parameters, although it may cause displacement of the endemic equilibrium.

In this paper, we have conducted a qualitative analysis of the impact of media coverage on the transmission of infectious diseases. We have provided a more realistic and more natural description for the epidemic model with media coverage by considering the real procedure of media propagation and education in disease transmission. The results obtained here could be beneficial for accurately assessing the effect of the media coverage in the control and treatment of infective diseases.

Acknowledgments

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Appendix. Brief introduction to the threshold policy

A threshold policy (TP) implies that, if the value of the threshold function is below the specified threshold level, control is suppressed, while if it is above the specified threshold level, control is implemented. As a result, a system subject to a TP leads to a variable structure system. The TP is also referred to as an on–off control in the control literature (see [17,18]). In the following, we shall establish a mathematical form for a TP. To illustrate our ideas, we choose a planar system as an example. A two-dimensional system submitted to a TP can be rewritten as

$$\frac{dx(t)}{dt} = f(x, u_\tau), \quad (\text{A.1})$$

where x ($x \in \mathbb{R}^2$) is the state vector, $f \in C(\mathbb{R}^2, \mathbb{R}^2)$ is continuous, and the control u_τ ($u_\tau \in \mathbb{R}$) is defined as

$$u_\tau = \begin{cases} u_1(x), & \tau(x) < 0, \\ u_2(x), & \tau(x) > 0, \end{cases} \quad (\text{A.2})$$

where $u_1(x) = 0$, $\tau(x)$ is a threshold dependent on the state vector x , and u_τ is not continuous. For the case when $\tau(x) = 0$, u_τ is not defined, i.e., u_τ is discontinuous on the boundary $\tau(x) = 0$. A ‘controlled system’ refers to one in which the control $u_\tau = u_2$ is implemented, and a ‘free system’ is one in which the control is not implemented (i.e., $u_\tau = 0$). Let

$$G^1 = \{x \in \mathbb{R}^2 | \tau(x) < 0\}, \quad G^2 = \{x \in \mathbb{R}^2 | \tau(x) > 0\},$$

and

$$\Sigma = \{x \in \mathbb{R}^2 | \tau(x) = 0\}. \quad (\text{A.3})$$

Denote the right-hand side of (A.1) (i.e., $f(x, u_\tau)$) in the region G^i by $f^{(i)}(x, u_\tau)$.

Definition A.1. Let x^* be such that $f^{(i)}(x^*, u_i) = 0$ for some u_i in (A.2). Then x^* is called a real equilibrium of system (A.1) with (A.2) if it belongs to G^i , and a virtual equilibrium if it belongs to G^j , $j \neq i$.

A ‘sliding mode’ may occur if there are subsets Σ_S of the manifold Σ such that the vectors $f^{(1)}(x, 0)$ and $f^{(2)}(x, u_2)$ are directed towards each other on them.

Definition A.2. A point x^* is said to be a pseudo-equilibrium of system (A.1) with (A.2) if it is an equilibrium of the sliding mode of system (A.1) with (A.2), i.e., $\lambda f(x^*, u_1(x^*)) + (1 - \lambda)f(x^*, u_2(x^*)) = 0$, $\tau(x^*) = 0$, where

$$\lambda = \frac{\langle \tau_x(x), f_x(x, u_2(x)) \rangle}{\langle \tau_x(x), f_x(x, u_2(x)) - f_x(x, u_1(x)) \rangle}.$$

Definition A.3. A point x^* is called a tangent point of system (A.1) with (A.2) if $x^* \in \Sigma_S$ and $\langle \tau_x(x^*), f^{(1)}(x^*, 0) \rangle = 0$ or $\langle \tau_x(x^*), f^{(2)}(x^*, u_2) \rangle = 0$.

Definition A.4. A point x^* is called a boundary equilibrium point of system (A.1) with (A.2) if $f^{(1)}(x^*, 0) = 0$, $\tau(x^*) = 0$ or $f^{(2)}(x^*, u_2) = 0$, $\tau(x^*) = 0$.

References

- [1] R.S. Liu, J.H. Wu, H.P. Zhu, Media/psychological impact on multiple outbreaks of emerging infectious diseases, *Comput. Math. Methods Med.* 8 (3) (2007) 153–164.
- [2] J.M. Tchuente, N. Dube, C.P. Bhunu, R.J. Smith, C. Bauch, The impact of media coverage on the transmission dynamics of human influenza, *BMC Public Health* 11 (Suppl. 1) (2011) S5.
- [3] J.A. Cui, Y.H. Sun, H.P. Zhu, The impact of media on the control of infectious diseases, *J. Dynam. Differential Equations* 20 (2008) 31–53.
- [4] J.A. Cui, X. Tao, H.P. Zhu, An SIS infection model incorporating media coverage, *Rocky Mountain J. Math.* 38 (5) (2008) 1323–1334.
- [5] C.J. Sun, W. Yang, J. Arino, K. Khan, Effect of media-induced social distancing on disease transmission in a two patch setting, *Math. Biosci.* (2011). <http://dx.doi.org/10.1016/j.mbs.2011.01.005>.
- [6] Y.F. Li, J.A. Cui, The effect of constant and pulse vaccination on SIS epidemic models incorporating media coverage, *Commun. Nonlinear Sci. Numer. Simul.* 14 (2009) 2353–2365.
- [7] S.Y. Tang, Y.N. Xiao, et al., Community-based measures for mitigating the 2009 H1N1 pandemic in China, *PLoS One* 5 (6) (2010) e10911. <http://dx.doi.org/10.1371/journal.pone.0010911>.
- [8] L.M. Cai, X.Z. Li, Analysis of a SEIV epidemic model with a nonlinear incidence rate, *Appl. Math. Model.* 33 (2009) 2919–2926.
- [9] F. Maggì, New flu has been around for years in pigs: study, *Reuters: Science* (2009) 11 June.
- [10] H.W. Hethcote, Qualitative analyses of communicable disease models, *Math. Biosci.* 28 (1976) 335–356.
- [11] M.J. Keeling, P. Rohani, *Modeling Infectious Diseases in Human and Animals*, Princeton University Press, Princeton, 2008.
- [12] A.F. Filippov, *Differential Equations with Discontinuous Righthand Sides*, Kluwer Academic, Dordrecht, 1988.
- [13] A.I. Reppasa, A.C. Tsoumanis, C.I. Siettos, Coarse-grained bifurcation analysis and detection of criticalities of an individual-based epidemiological network model with infection control, *Appl. Math. Model.* 34 (2010) 552–560.
- [14] Y.N. Xiao, Y.C. Zhou, S.Y. Tang, Modelling disease spread in dispersal networks at two levels, *IMA J. Math. Appl. Med. Biol.* 28 (2011) 227–244.
- [15] S.L. Sun, L.S. Chen, Mathematical modelling to control a pest population by infected pests, *Appl. Math. Model.* 33 (2009) 2864–2873.
- [16] R.M. Corless, G.H. Gonnet, et al., On the Lambert W function, *Adv. Comput. Math.* 5 (1996) 329–359.
- [17] V.I. Utkin, *Sliding Modes and their Applications in Variable Structure Systems*, Mir, Moscow, 1978.
- [18] V.I. Utkin, *Sliding Modes in Control and Optimization*, Springer, Berlin, 1992.
- [19] Yu.A. Kuznetsov, S. Rinaldi, A. Gragnani, One parameter bifurcations in planar Filippov systems, *Internat. J. Bifur. Chaos Appl. Sci. Engrg.* 13 (2003) 2157–2188.
- [20] M.D. Bernardo, C.J. Budd, A.R. Champneys, et al., Bifurcations in nonsmooth dynamical systems, *SIAM Rev.* 50 (2008) 629–701.
- [21] W.D. Wang, Backward bifurcation of an epidemic model with treatment, *Math. Biosci.* 201 (2006) 58–71.
- [22] Y.N. Xiao, X.X. Xu, S.Y. Tang, Sliding mode control of outbreaks of emerging infectious diseases, *Bull. Math. Biol.* 74 (2012) 2403–2422.
- [23] M.I.S. Costa, E. Kaszkurewicz, A. Bhaya, L. Hsu, Achieving global convergence to an equilibrium population in predator–prey systems by the use of a discontinuous harvesting policy, *Ecol. Model.* 128 (2000) 89–99.
- [24] M.E.M. Meza, M.I.S. Costa, A. Bhaya, E. Kaszkurewicz, Threshold policies in the control of predator–prey models, in: *Preprints of the 15th Triennial World Congress, IFAC, Barcelona, Spain, 2002*, pp. 1–6.
- [25] S.Y. Tang, J.H. Liang, Y.N. Xiao, R.A. Cheke, Sliding bifurcation of Filippov two stage pest control models with economic thresholds, *SIAM J. Appl. Math.* 72 (2012) 1061–1080.
- [26] S.Y. Tang, J.H. Liang, Global qualitative analysis of a non-smooth Gause predator–prey model with a refuge, *Nonlinear Analysis RWA* 76 (2013) 165–180.
- [27] M.E.M. Meza, A. Bhaya, E.K. Kaszkurewicz, D.A. Silveira, M.I. Costa, Threshold policies control for predator–prey systems using a control Liapunov function approach, *Theor. Popul. Biol.* 67 (2005) 273–284.
- [28] D.S. Boukal, V. Krivan, Lyapunov functions for Lotka–Volterra predator–prey models with optimal foraging behavior, *J. Math. Biol.* 39 (1999) 493–517.